Predicting Purchasing Behavior at PriceMart

Parts A & B — From Linear Regression to Regularized Logistic Models

Predictive modeling for the Mom & Baby unit | **Analysts: Mark Morse & Jill Wehunt**

Submission By: Group 6

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Agenda

01 **Business Context & Objective**
Why PriceMart wants to predict pregnancy from purchase data

02 **Building the Dataset**
Features, dummy coding, and design trade-offs

03 **Training & Validation Design**
Train/holdout split and 5-fold cross-validation

04 **Linear vs. Logistic Regression**
Model mechanics, MLE, and Part A results

05 **Regularization — Ridge & Lasso**
Tackling overfitting in Part B

06 **Model Comparison & Feature Impact**
Log loss, ROC, confusion matrices

07 **From Predictions to Payoff**
Cutoffs, the payoff matrix, and expected value

08 **Key Takeaways & Recommendation**
What PriceMart should actually do

Why PriceMart Went Looking for Pregnancy Signals

- PriceMart is a national retailer; its Mom & Baby unit sits inside personal care & beauty.
- VP of Marketing Jill Wehunt observed that new parents form shopping habits fast — households buying diapers online kept buying Mom & Baby products for 18+ months.
- Hypothesis: if PriceMart earns a household's business before the baby arrives, it can lock in years of loyalty.
- Analyst Mark Morse was tasked with turning this hunch into a data-driven targeting model.

THE CORE QUESTION

"Can we predict which customer households are newly pregnant or trying to conceive — using only their purchase data?"

Goal: flag likely-pregnant households early so digital marketing can send targeted promotions before competitors do.

The Feature Set — Table 1: Taxonomy

From millions of transaction records, Morse's team selected 17 independent features believed to signal pregnancy, plus one binary target.

Demographic (Categorical)	Purchase (Binary 0/1)	Purchase (Quantity)	Stop-Purchase (Binary 0/1)	Target Feature (Binary 0/1)
• Gender • Housing type	• Pregnancy test • Birth control • Feminine hygiene • Folic acid • Prenatal yoga • Maternity clothes • Body pillow • Ginger ale • Sea bands • Cigarettes • Wine	• Vitamins (unit count)	• Stopped buying cigarettes • Smoking cessation (patch) • Stopped buying wine	• Pregnant (1) / • Not Pregnant (0)

Preparing the Data — Dummy Coding (Exhibit 1)

Categorical fields (gender, housing type) had to become numeric before the regression could use them. A category with k levels needs k−1 dummy columns.

Original Data Format

Implied Gender	Home/Apt/PO Box
M	A
M	H
M	H
F	H
F	A
F	H
M	H



dummy coding

Dummy Coding Format

Male	Female	Home	Apt
1	0	0	1
1	0	1	0
1	0	1	0
0	1	1	0
0	1	0	1
0	1	1	0
1	0	1	0

Rule of thumb

3 categories (Home / Apt / PO Box) → 2 dummy columns.

If both dummies = 0 → implied category is PO Box.

Same logic applied to Male/Female (with 'Unknown' as the implied 3rd category).

Design Trade-off: Binary Flag or Quantity?

Every feature also required a judgment call: capture whether a household bought a product, or how much? Morse's team weighed this per product.

Vitamins → Quantity

Coded as a count, not a flag. Hypothesis: women trying to conceive take noticeably larger quantities of vitamins than their normal baseline — the magnitude itself carries signal.

Ginger Ale → Binary (0/1)

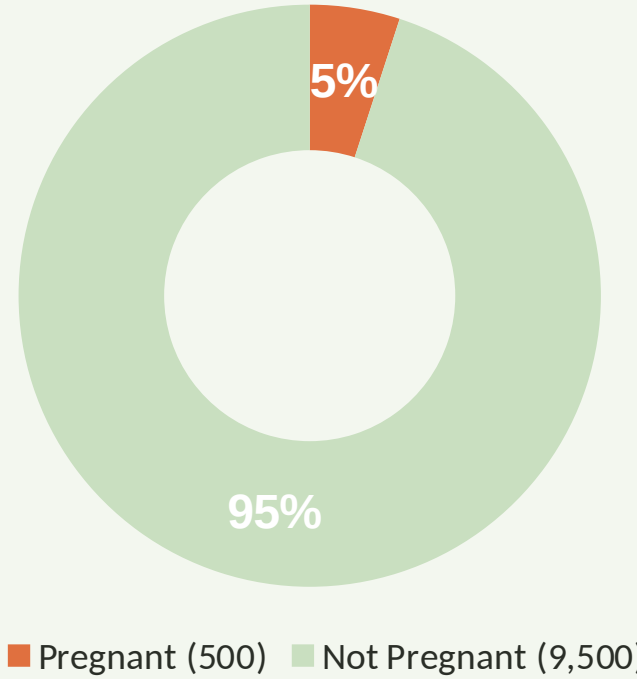
Ginger ale eases early-pregnancy nausea. The team debated quantity vs. presence, and ultimately treated it as a simple purchase/no-purchase flag — whether a household is buying it at all, not how much.

The open question

Morse had no fixed rule for when to use quantity vs. binary — and worried about what to do when the team simply wasn't sure. This ambiguity is a real limitation of the feature-engineering process.

A Rare-Event Problem: Severe Class Imbalance

Cleaned dataset: 10,000 household records



5%

of households in the data are pregnant — exactly the class PriceMart most wants to identify.

Why it matters: a model trained on this mix will naturally lean toward predicting "not pregnant" — it can be 95% accurate by ignoring the minority class entirely. Every downstream design choice (balanced vs. unbalanced training, regularization, cutoff selection) exists to counter this.

Step 1: Training Set vs. Holdout Set (Exhibit 3a)

Full data set: 10,000 households (9,500 not-pregnant · 500 pregnant)



TRAINING SET · 80%

8,000 households

400 pregnant + 7,600 not-pregnant

HOLDOUT · 20%

2,000
100 preg. +
1,900 not

Why hold data out?

The holdout set is never touched during model building. It's the only honest test of how the model performs on data it has never seen.

5 cross-validation folds (20% each)

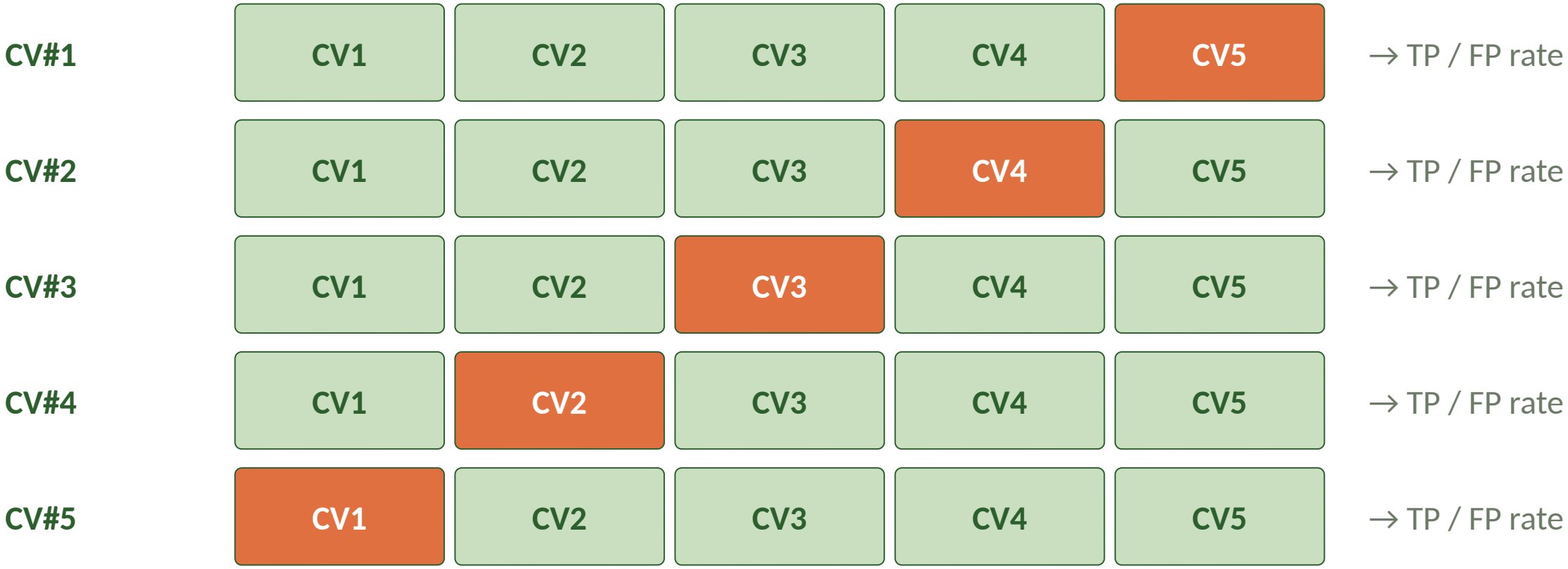
Each fold: 80 pregnant + 1,520 not-pregnant = 1,600 households — used to train and cross-test the model multiple times.

Step 2: 5-Fold Cross-Validation (Exhibit 4)

Used to train

Held out for validation

The training set (subsets 1–5) rotates which fold is held back for validation, so every fold serves as the test set exactly once.



Legend:

Average of 5 validation scores → one performance estimate per model

Open Question: Balanced or Unbalanced Training? (Exhibit 3b)

Unbalanced Approach

Use all 8,000 training households as-is:
400 pregnant + 7,600 not-pregnant

✓ Reflects the real-world population the model will actually see

× Model may skew toward predicting "not pregnant," since that class dominates the loss function

Balanced Approach

Downsample to equal classes:
400 pregnant + 400 not-pregnant

✓ Model focuses equally on both classes, better learns pregnancy signal

× No longer representative of the true population — may bias probability estimates

Resolving It: Keep the Natural 5% / 95% Mix

Recommendation: Train and validate on the real, unbalanced population (400 pregnant / 7,600 not-pregnant) — don't downsample to 50/50. Let regularization and the payoff-driven cutoff do the work that balancing was meant to do.

1 The case's own exhibits already decide it

Exhibit 4's cross-validation folds hold 80 pregnant / 1,520 not-pregnant each — the true ~5% mix, not a rebalanced 50/50 split. Every log loss figure in Table 1 (0.1514 → 0.1280) was computed on that unbalanced structure.

2 Calibration beats rebalancing

Log loss rewards well-calibrated probabilities. Training on an artificial 50/50 set inflates predicted pregnancy probabilities relative to the real 5% base rate — distorting the exact numbers the payoff-matrix step later depends on.

3 Regularization already treats the real risk

Balancing was floated to fight overfitting from only 400 positive cases. Ridge regression solves that same variance problem directly, without touching the class mix or biasing the probabilities.

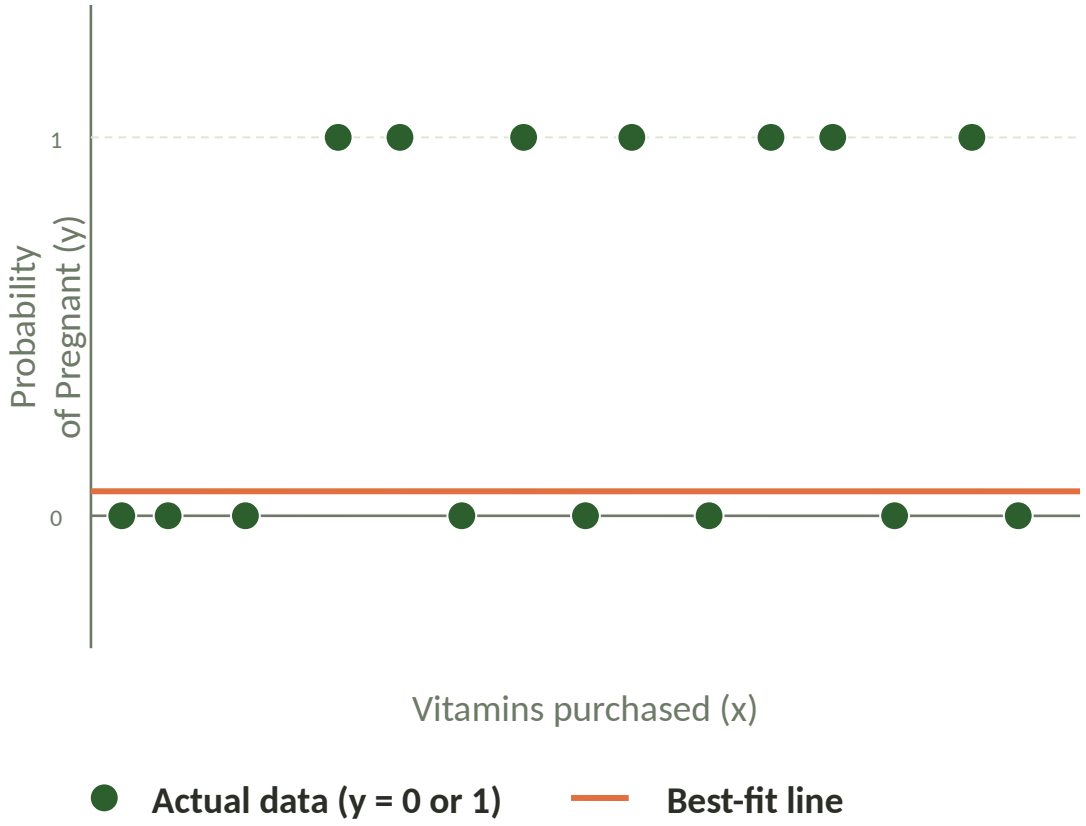
4 The cutoff, not the training mix, is the right lever

PriceMart already has a purpose-built tool for the rare-event problem: a business-driven cutoff chosen by expected payoff (Tables 4–5). That's a more precise fix than blunt rebalancing.

Bottom line: rebalancing is a reasonable technique in general — but here it would work against the very metric (log loss) and payoff calculation the team relies on later. Leave the mix alone; let regularization and cutoff selection carry that load instead.

Starting Point: Linear Regression Breaks Down

Morse first fit $y = \beta_0 + \beta_1 x$ on a single feature (vitamins purchased) to predict pregnancy status (Appendix, Figure A-1).



The core flaw

The fitted line **y = 0.0918 + 0.0136x** can produce predictions **above 1 or below 0** — meaningless for a probability.

$R^2 = 0.095$, F-statistic = 104.8 ($p < 0.01$) → statistically significant relationship, but a poor overall fit — vitamins alone explain under 10% of the variation.

This motivated a model whose output is mathematically constrained to [0, 1] — logistic regression.

Logistic Regression: Bounding Predictions to [0, 1]

The Sigmoid Link Function

$$y = f(z) = 1 / (1 + e^{-z})$$

z = -5 → f(z) ≈ 0.0067 (near 0)
z = +5 → f(z) ≈ 0.99 (near 1)

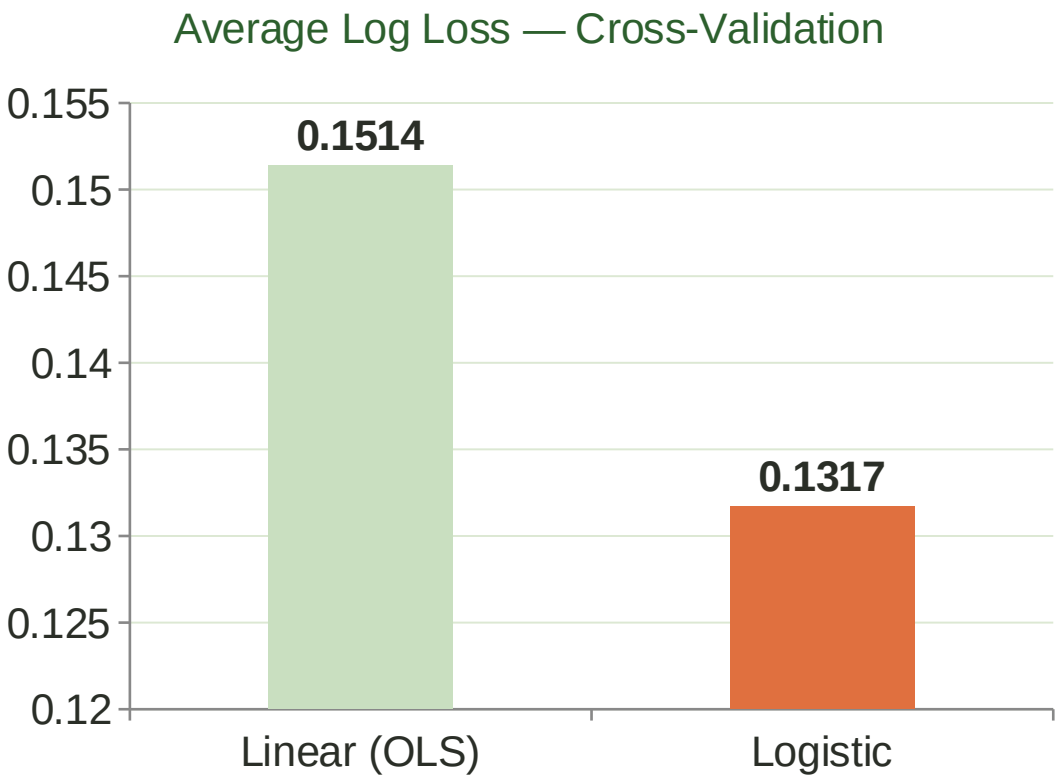
No matter how extreme the inputs, the output always lands between 0 and 1 — a valid probability of pregnancy.

How Coefficients Are Estimated: MLE

Maximum Likelihood Estimation chooses $\beta_0 \ldots \beta_n$ to maximize $p^y(1-p)^{(1-y)}$ across all observations — i.e., predicted probabilities that match actual outcomes as closely as possible.
For computational ease, this becomes maximizing the **log-likelihood** — equivalently, minimizing **log loss** (negative sum of log-likelihoods).

This replaces the Sum of Squared Errors (SSE) used in linear regression — SSE is not the right objective once predictions are probabilities.

Result: Logistic Regression Wins Part A (Exhibit 6)



Lower log loss = better-calibrated probabilities

What Exhibit 6 shows

13.0% lower log loss for logistic vs. linear regression on the same cross-validation folds.

Feature significance mostly agrees: variables significant in linear regression stayed significant in logistic regression — except ginger ale, significant in linear but not in logistic at the 5% level.

Signs (direction) of coefficients matched across both models — e.g., pregnancy test, folic acid, prenatal yoga, maternity clothes all push the probability of pregnancy up; birth control, cigarettes, wine push it down.

Logistic regression also lacks R^2, F- and t-tests — Morse instead used pseudo-R^2 and the Wald statistic.

Insight: Bounded, probability-consistent output — not fit alone — is why logistic regression was the better choice.

Exhibit 6 — Linear vs. Logistic Regression Coefficients

Log loss: **OLS 0.1514** vs. **Logistic 0.1317** — coefficient, then (p-value); **bold navy = significant at 5%**, *muted italic = not significant*

Feature	OLS β (p)	Logistic β (p)
Constant	*0.0700 (0.492)*	**-4.3631 (0.000)**
Apartment	*-0.0071 (0.275)*	*-0.1723 (0.379)*
Birth Control	**-0.0450 (0.000)**	**-2.5745 (0.000)**
Body Pillow	*0.0119 (0.407)*	*-0.1684 (0.623)*
Cigarettes	**-0.0287 (0.000)**	**-1.7962 (0.000)**
Female	*0.0036 (0.558)*	*-0.0008 (0.996)*
Feminine Hygiene	**-0.0453 (0.000)**	**-2.1206 (0.000)**
Folic Acid	**0.4244 (0.000)**	**2.5175 (0.000)**
Ginger Ale	**0.0371 (0.000)**	*0.3013 (0.120)*
Home	*0.0036 (0.584)*	*0.2211 (0.252)*
Ice Cream	*-0.0001 (0.182)*	*-0.0021 (0.372)*

Feature	OLS β (p)	Logistic β (p)
Male	*-0.0031 (0.617)*	*-0.1670 (0.359)*
Maternity Clothes	**0.0542 (0.000)**	**1.0664 (0.000)**
Pregnancy Test	**0.1128 (0.000)**	**1.4193 (0.000)**
Vitamins	**0.0022 (0.000)**	**0.0636 (0.000)**
Prenatal Yoga	**0.2079 (0.000)**	**3.0268 (0.000)**
Sea Bands	**0.0317 (0.007)**	**0.8309 (0.002)**
Smoking Cessation	**0.0858 (0.000)**	**1.5327 (0.000)**
Stopped Buying Cigarettes	**0.0267 (0.001)**	**0.3280 (0.045)**
Stopped Buying Wine	**0.0330 (0.000)**	**0.5452 (0.000)**
Wine	**-0.0341 (0.000)**	**-2.2697 (0.000)**

PART B

The Overfitting Problem — Why Regularize?

Even after choosing logistic regression, Morse remained concerned the model would **overfit the training data** — fitting noise and quirks specific to these 8,000 households rather than the true underlying pattern.

1	**17 features, only 500 positive cases**	High variance relative to sample size — a classic overfitting setup
2	**Coefficients may chase noise**	Large, unstable beta weights fit training idiosyncrasies, not general signal
3	**Poor generalization**	A model that memorizes training data predicts poorly on genuinely new customers

Solution: constrain the size of the beta coefficients so the model can't fit the training data too perfectly → Ridge and Lasso regression.

Two Ways to Shrink the Coefficients

Ridge Regression (L2)

$$\beta_0^2 + \beta_1^2 + \ldots + \beta_n^2 \leq t$$

Adds a penalty (λ) on the **sum of squared** coefficients, subtracted from the log-likelihood.

Shrinks all coefficients toward zero, but rarely to exactly zero — every feature stays in the model, just down-weighted.

Especially useful when variance is high — e.g. small sample size or many features relative to observations (exactly PriceMart's situation).

λ is tuned by testing multiple values and picking the one that minimizes average log loss across the 5 CV folds.

Lasso Regression (L1)

$$|\beta_0| + |\beta_1| + \ldots + |\beta_n| \leq t$$

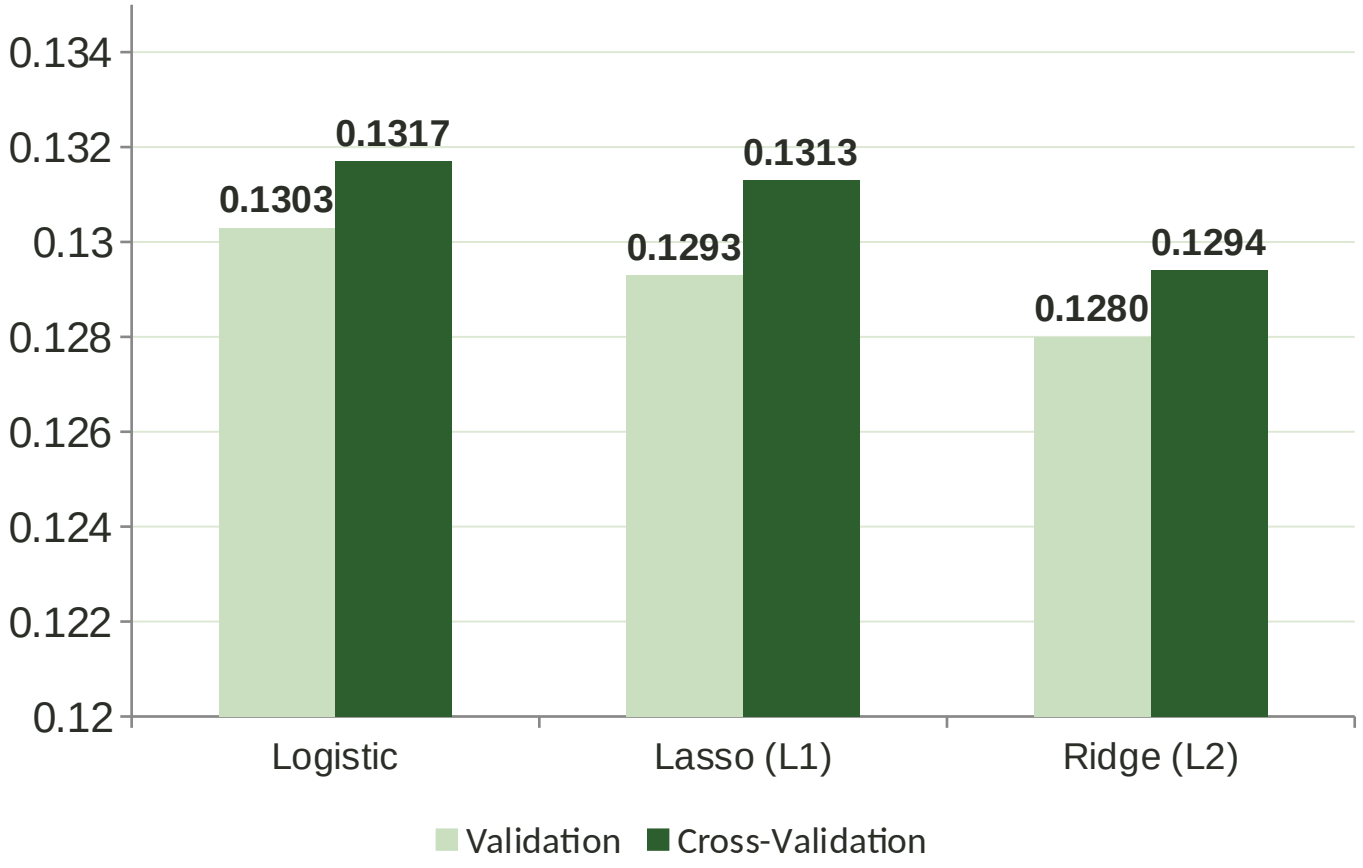
Adds a penalty (λ) on the **sum of absolute values** of coefficients.

Can force weak coefficients to exactly zero — effectively performing automatic feature selection and producing a simpler model.

Same tuning process as ridge: sweep λ, cross-validate, choose the value minimizing average log loss.

Trade-off: a simpler model, but one that discards information some analysts might still find relevant.

Three Models, Head-to-Head — Average Log Loss (Table 1)



Reading the result

Ridge wins — lowest log loss on both validation (0.1280) and cross-validation (0.1294).

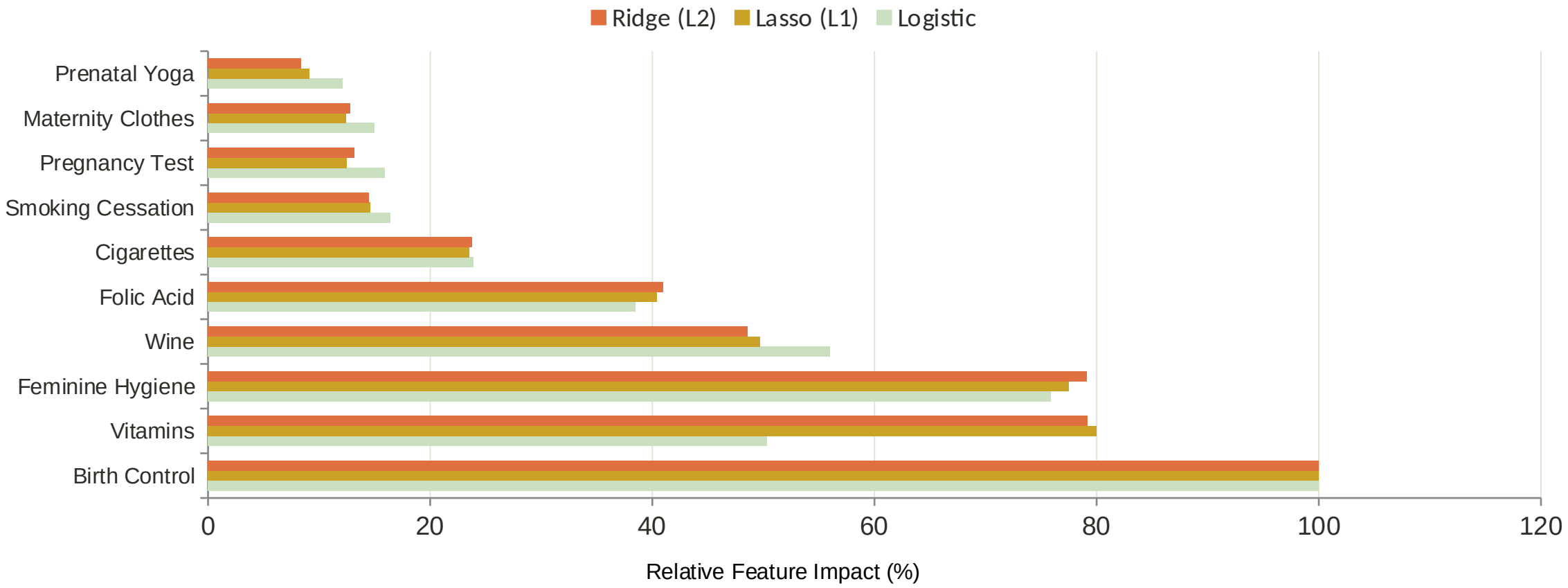
Lasso is close behind, edging out plain logistic — some feature simplification helps without much cost.

Plain logistic performs worst of the three — confirms the overfitting concern was valid.

But the gaps are small (≈0.003 log loss) — all three models are directionally similar; the choice matters more for interpretability and stability than for raw accuracy.

Which Features Matter Most? (Exhibit 1, Part B)

Feature impact, normalized to the strongest predictor (Birth Control = 100%) — compared across all three models.



Why Ridge Was the Right Call for PriceMart

1 Lowest log loss

0.1280 on validation and 0.1294 on cross-validation — best of the three models on the metric Morse chose to optimize.

2 Keeps all features

Unlike Lasso, Ridge doesn't zero out coefficients — useful when the business wants to understand every signal's directional effect, not just the strongest ones.

3 Feature ranking is stable

Birth control, vitamins, and feminine hygiene dominate impact in all three models — the story doesn't change much across methods, building confidence in the signal.

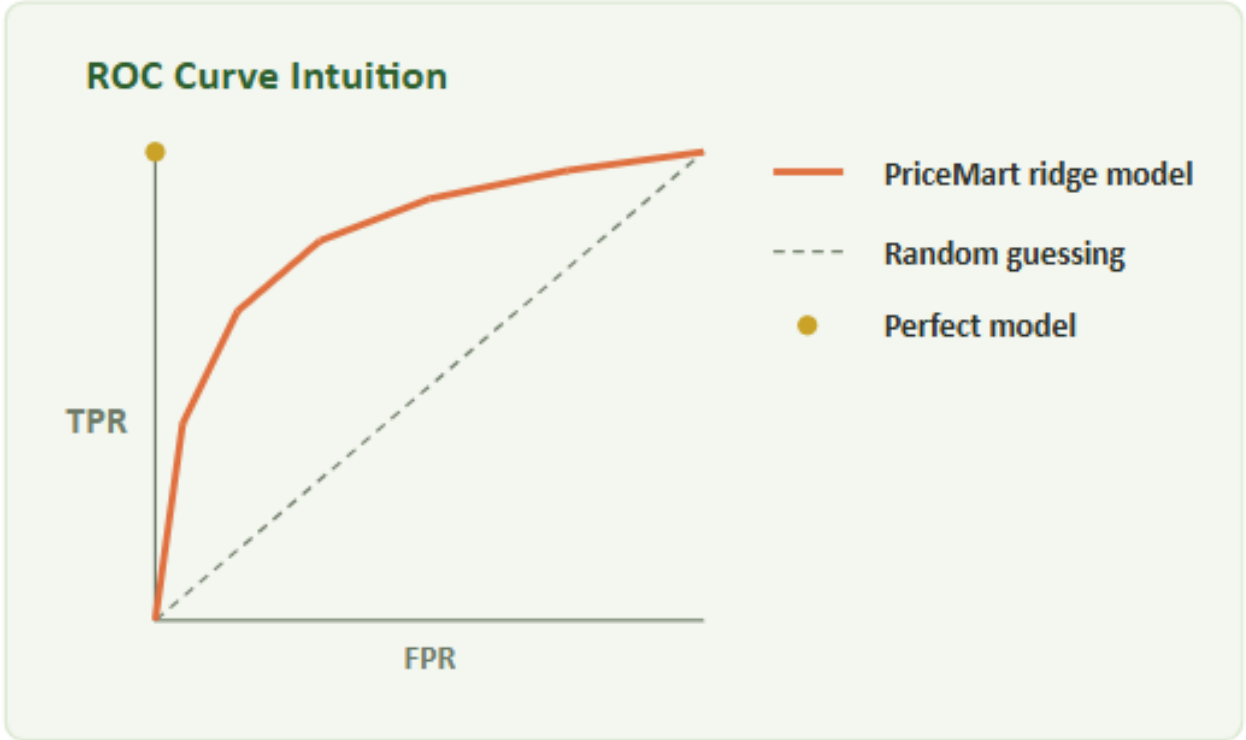
4 Right tool for high variance

With 17 features and only 400 positive training cases, ridge's variance-reduction is exactly the regularization this small, imbalanced dataset needed.

Turning Probabilities into Decisions: Cutoffs & the Confusion Matrix

The ridge model outputs a probability per household. PriceMart still has to pick a cutoff — above it, target the household as "pregnant." Every cutoff creates a trade-off between catching true positives and provoking false positives.

Predicted → \ Actual →	Pregnant	Not Pregnant
Pregnant	**TRUE POSITIVE (A)** Predicted pregnant, actually pregnant ✓	**FALSE POSITIVE (Y)** Predicted pregnant, actually, not pregnant
Not Pregnant	**FALSE NEGATIVE (Z)** Predicted not pregnant, actually pregnant	**TRUE NEGATIVE (B)** Predicted not pregnant, actually, not pregnant ✓



Rank households by predicted probability and sweep the cutoff from high to low. The steeper and more top-left the curve, the better the model separates pregnant from non-pregnant households.

Same Model, Three Cutoffs — Very Different Outcomes

Applied to the 8,000-household training set (400 pregnant / 7,600 not-pregnant). As the cutoff drops, the model casts a wider net.

Cutoff = 0.727

	Pregnant	Not Preg.
Pred. Pregnant	31 (7.75%)	0 (0%)
Pred. Not Preg.	369 (92.25%)	7,600 (100%)

Very conservative — almost no false alarms, but misses 92% of pregnant households.

Cutoff = 0.40

	Pregnant	Not Preg.
Pred. Pregnant	100 (25%)	44 (0.57%)
Pred. Not Preg.	300 (75%)	7,556 (99.42%)

Balanced middle ground — catches 1 in 4 pregnant households at a low false-alarm rate.

Cutoff = 0.10

	Pregnant	Not Preg.
Pred. Pregnant	289 (72.25%)	930 (12.23%)
Pred. Not Preg.	111 (27.75%)	6,670 (87.76%)

Aggressive — catches nearly 3 in 4 pregnant households, but flags 930 non-pregnant ones too.

Assigning Business Value: The Payoff Matrix (Table 4)

Wehunt and Morse assigned a dollar value to each outcome — turning a purely statistical model into a business decision tool.

Value framing

A successful catch (+ $500) is worth 5x more than the cost of a wasted promotion (– $100). This asymmetry matters a lot for the optimal cutoff.

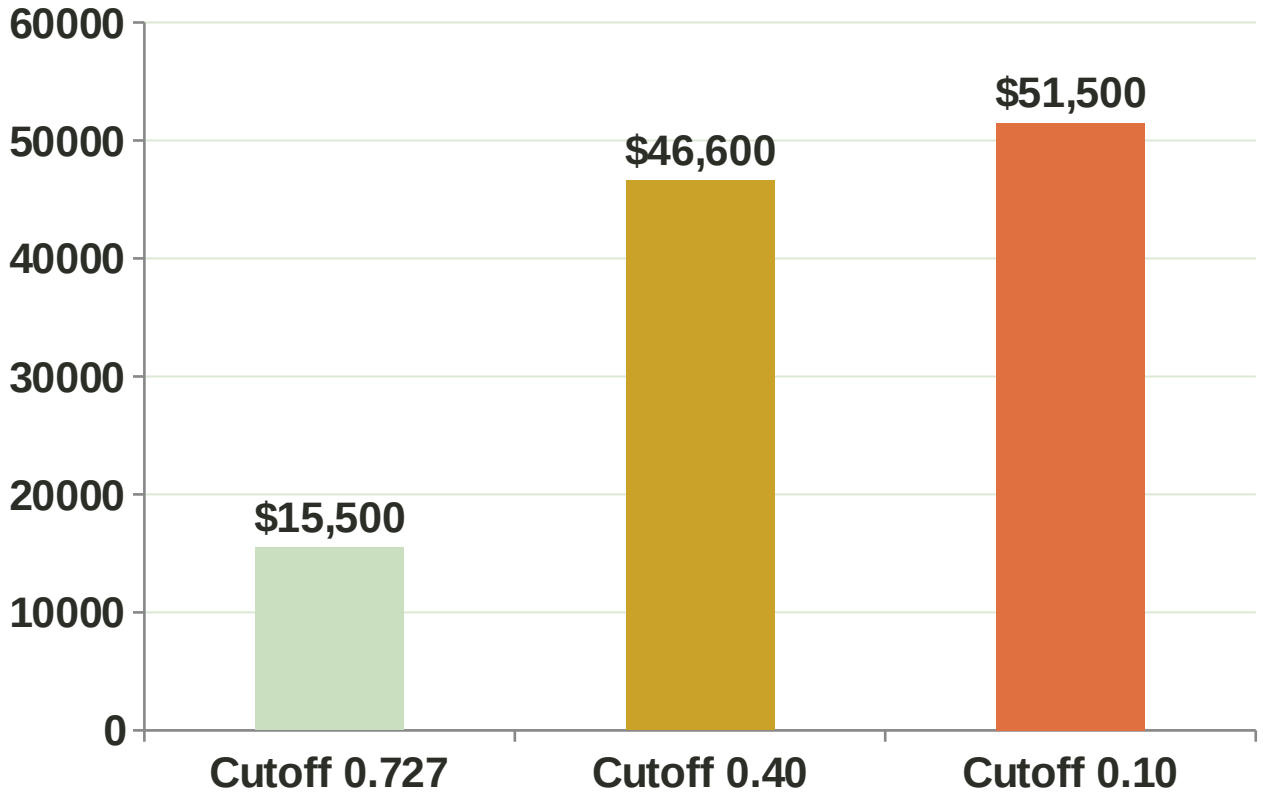
PREDICT PREGNANT **Actually Pregnant** **+$500**	**PREDICT PREGNANT** **actually, Not Pregnant** **–$100**
PREDICT NOT PREGNANT **Actually Pregnant** **$0**	**PREDICT NOT PREGNANT** **Actually Not Pregnant** **$0**

Why this matters

With false positives 5x cheaper than true positives are valuable, the profit-maximizing cutoff is not necessarily the one that looks most "balanced" on a confusion matrix. Business value, not just classification accuracy, should drive the cutoff choice — this is why the payoff matrix step exists.

Expected Payoff by Cutoff (Table 5)

Expected payoff = (A×$500) + (Y×−$100) + (Z×$0) + (B×$0), summed across all households in the training set.



The lower the cutoff, the higher the payoff

Because a true positive ($500) is worth 5x more than a false positive costs (−$100), casting a wider net — even at the price of many false alarms — maximizes expected value. Cutoff 0.10 outperforms 0.727 by more than 3x.

Key Takeaways

1. Rare-event, imbalanced data (5% positive class) shapes every modeling decision — from balanced training to cutoff selection.

2. Logistic regression beat linear regression because bounded, probability-consistent output matters more than raw fit (13% lower log loss).

3. Regularization (Ridge) beat plain logistic regression by controlling variance — the lowest log loss of all three models tested.

4. Feature rankings were consistent across models — birth control, vitamins, and feminine hygiene are the strongest, most trustworthy signals.

5. The right cutoff is a a business decision, not a statistical one — it should maximize expected payoff, not accuracy or a "balanced" confusion matrix.

6. Given the case's own payoff asymmetry ($500 gain vs. $100 cost), a low cutoff (~0.10) delivers the highest expected value — over 3x the conservative cutoff.

RECOMMENDATION

Deploy the Ridge Logistic Model — Set the Cutoff by Expected Value, Not Instinct

- Use the L2-regularized (ridge) logistic model — it had the lowest log loss and preserves all 17 features for interpretability.
- Set the targeting cutoff using the payoff matrix, not a default 0.5 threshold — the case's own numbers favor a low cutoff near 0.10.
- Revisit the payoff assumptions ($500 / −$100 / $0 / $0) with real cost and lifetime-value data before finalizing — they were estimates, and false-negative cost of $0 likely understates the true opportunity loss.
- Retrain and re-validate periodically as new purchase data accumulates — the model's stability should be monitored, not assumed.

PREDICTING PURCHASING BEHAVIOR AT PRICEMART

Thank You

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